

Identification of Predictive Subphenotypes for Clinical Outcomes Using Real World Data and Machine Learning

Corresponding Author: Professor Fei Wang

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The paper details the use of a proposed computational framework, the Graph-Encoded Mixture Survival model (GEMS), to identify groups of patients with coherent time-to-event survival patterns using EHR data. The variables included are demographics, lab tests, vital signs, diagnoses, metastasis status, and medications. Through system optimization, three groups of patients (defined as subphenotypes) were discovered. Statistical analysis and data visualization of the top features for each patient group demonstrate their significance and relevance to the survival of the non-small cell lung cancer (NSCLC) patient cohort. The paper is well-written and provides extensive details about the computational framework, a comparison with baseline methods, and an ablation study evaluating the impact of each module.

Some questions need to be addressed by the authors:

1. Patient Graph Construction: The patient graph was built using the five most similar patients. There is no information on how this number was determined. Why not use a similarity threshold instead of a fixed number of five? Some patients might not have five very similar neighbors.
2. Selection of comorbidities and lab tests: The authors state, "Evaluation of comorbidities and metastasis was grounded on all documented International Classification of Diseases (ICD-10 and ICD-9) diagnoses within the 445 observation period." However, only four features are listed in Supplemental Table 5. Please explain how these diseases were selected. Similarly, only a selected set of lab tests was included in the study. Please clarify the selection criteria for these lab tests.
3. Missing Data Imputation: For the imputation of missing data, clarify whether the mean values were calculated based on the training data and then used to impute both the training and validation sets.
4. Comparison of Methods: Table 2 reports the comparison of methods using the validation set. Please also include a comparison of methods using the training set to demonstrate that there is no overfitting issue.
5. Vital Signs in Figure 4: Group C is described as including abnormal lab tests and vital signs, yet no vital signs appear to be included. Does this mean vital signs did not contribute to the predictive model? Please explain. Additionally, abnormal lab tests can indicate either high or low values. It would be beneficial to analyze whether each abnormality is high or low for each lab test in this figure.
6. Exclusion of Age: Age was not included in the list of demographic factors that were input into the model. Please provide an explanation for its exclusion.

Some minor comments:

1. The limitations section should address whether the proposed framework can be applied to other disease analyses.
2. Although the framework used the last lab test results before the index date, a discussion on the potential benefits of considering changes in lab test results for NSCLC or other diseases would be useful.

(Remarks on code availability)

Based on the information provided, the code pertains to a different manuscript titled "Generalizability of a Machine Learning Model for Improving Utilization of Parathyroid Hormone-Related Peptide Testing Across Multiple Clinical Centers." Although some portions of the code may be applicable to this submission, it would enhance reproducibility to create a separate repository specifically dedicated to this manuscript.

Reviewer #3

(Remarks to the Author)

The authors developed a framework to identify predictive subphenotypes of advanced non-small cell lung cancer (aNSCLC), a heterogeneous disease, from electronic health records (EHR). The subphenotypes are grouped based on baseline clinical characteristics and coherent and distinct time-to-event overall survivals, and the EHR data includes 104 values from demographics, lab tests, vital signs, etc. The framework consists of graph neural network (GNN) encoder, clustering, and a mixture survival predictor. It was trained retrospectively on 3225 patients from the ConcertAI Patient360 NSCLC dataset and validated on 1441 patients from the same database but from a separate region. Hyperparameter tuning, including the number of subphenotypes, was performed on the development cohort and a model was developed to predict subphenotypes. This model was used on the validation cohort, and the phenotypes were validated for consistency. The feature importance was analyzed, and the C-index and pairwise log-rank scores were compared for the current framework against several alternative baseline models. The subphenotypes that were determined for aNSCLC have meaningful clinical characteristics, but causal conclusions cannot be drawn about the association between the EHR features and the treatment response.

This was a comprehensive, well-done study. The proposed model was compared to multiple baselines, the figures were informative and illustrated important concepts, figures 3 and 4 make it easy to see that the phenotypes are consistent across the development and validation set, and the methods described the cost function in detail and how the hyperparameters were selected so that the reader can be confident that you're not overhyping and an appropriate number of phenotypes were determined.

There are not any major concerns, but there are some additional details that would be nice to report.

1. I would suggest the use of Table 1 from Bos et. al.'s paper "The perils of premature phenotyping in COVID-19: a call for caution" as a checklist for mature phenotypes. I would check a box next to timing, derivation, size of derivation cohort, validation, internal coherence, relationship to underlying biology, and complexity. Impact on practice was mentioned as a limitation in the paper. Discreteness/continuity is the only row in this table that may need slightly more explanation in the discussion section. You describe how phenotype 2 is different than phenotypes 1 and 3, and you justify selecting three phenotypes based on majority voting from the results of NbClust, but it should be mentioned whether phenotype 2 is a discrete phenotype or is a continuous part of the space between phenotypes 1 and 3? It would also be informative to report the full results of NbClust; were three subphenotypes a clear winner or only slightly better than two subphenotypes?
2. On line 482, the graph neural network uses the top-5 nearest neighbors. Could this have been a hyperparameter? How can you justify that the proposed method is either insensitive to this choice or that this choice is optimal?
3. On line 539, it was stated that the lambda parameters could be determined empirically. Would you report the values of lambda that were used? It would be informative to know how much the loss function was weighted towards each individual component.

There are also a few typos to correct:

1. Line 517 says uncensored but you meant censored
2. Line 555 period should be a comma and We should be lowercase
3. Line 604 change my method to our proposed method
4. Supplementary table 3, three column headers have the word percentage split across two lines without a hyphen.

(Remarks on code availability)

Reviewer #4

(Remarks to the Author)

Pan et al present a graph neural network (GNN) based approach to phenotypically cluster patients with EHR data, called GEMS. They apply this approach to the concertAI database including patients with advanced NSCLC treated with immunotherapy and find, in both 5 fold cross validation and a held out cohort in the same database, that the clusters have prognostic power. However, the method is not benchmarked against more straightforward clustering methods aside from a qualitative UMAP presentation in figure 2; the justification for the GNN is not clear. The molecular data included seem somewhat arbitrary; several genes with known significance are excluded. The survival prediction results are benchmarked against standard methods but perform only marginally better if at all (confidence intervals are not provided, and the c indices are better than the next best methods by 1-2%). The important features identified by the model are already known to have prognostic significance in NSCLC. Although the analyses are reasonably performed, the results appear to be of limited computational and biological significance. I have some suggestions to improve the manuscript, but to be honest, unless the results of these suggested analyses produce something unexpectedly striking, the novelty and impact of this manuscript will likely still be quite limited.

-Straightforward clustering on the features directly without the GNN should be added as a method for benchmarking in the survival analyses.

-Full molecular data from tumor genomic profiling should be included. I see tumor mutational burden, MET, PTEN, and STK11 were included so this must exist for the cohort, but these 4 features seem extremely arbitrary/poorly chosen.

-The GEMS algorithm developed in the immunotherapy cohort should be applied to non-immunotherapy cohorts to test whether the clusters are predictive of differential response to immunotherapy or merely prognostic.
-Results from the individual fold cross validation results would help readers get a sense of the variation in c indices.
-It would be helpful to know what immunotherapy treatments patients received and in combination with what other cytotoxic or targeted agents.

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have thoroughly addressed my comments, revised the manuscript accordingly, and provided additional information in the supplemental materials.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

The authors addressed all my concerns, thank you.

(Remarks on code availability)

Code and example dataset were made available.

Reviewer #4

(Remarks to the Author)

The authors have made a reasonable attempt to address my comments given the limitations of their dataset.

(Remarks on code availability)

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Review Response Letter

We appreciate the constructive comments and feedback from the reviewers. Below please find our detailed point-by-point response for each of them.

Reviewer #1 (Remarks to the Author):

The paper details the use of a proposed computational framework, the Graph-Encoded Mixture Survival model (GEMS), to identify groups of patients with coherent time-to-event survival patterns using EHR data. The variables included are demographics, lab tests, vital signs, diagnoses, metastasis status, and medications. Through system optimization, three groups of patients (defined as subphenotypes) were discovered. Statistical analysis and data visualization of the top features for each patient group demonstrate their significance and relevance to the survival of the non-small cell lung cancer (NSCLC) patient cohort. The paper is well-written and provides extensive details about the computational framework, a comparison with baseline methods, and an ablation study evaluating the impact of each module. Some questions need to be addressed by the authors:

1. Patient Graph Construction: The patient graph was built using the five most similar patients. There is no information on how this number was determined. Why not use a similarity threshold instead of a fixed number of five? Some patients might not have five very similar neighbors.

Answer: Thanks for the question. We have added Supplementary Table S11 to compare different strategies by varying similarity thresholds and using fixed numbers of neighbors, based on performance in the development cohort. The results in the table show that using a fixed number of five neighbors yielded the best performance.

2. Selection of comorbidities and lab tests: The authors state, "Evaluation of comorbidities and metastasis was grounded on all documented International Classification of Diseases (ICD-10 and ICD-9) diagnoses within the 445 observation period." However, only four features are listed in Supplemental Table 5. Please explain how these diseases were selected. Similarly, only a selected set of lab tests was included in the study. Please clarify the selection criteria for these lab tests.

Answer: Thanks for the comment. The ConcertAI EHR database comprises structured and unstructured curated records derived from oncology clinics only, with majority sourced from community hospitals. As a result, records pertaining to comorbidities are limited either to what was previously reported to the oncology clinic or to the timeframe during which the patient was seen at the oncology clinic. This is a known limitation ([1]) for oncology clinic-sourced EHR databases. The diagnoses were derived by rolling from ICD-9/10 codes to comorbidity class provided in HCUP/AHRQ Elixhauser comorbidity index ([2]) and comorbidities classes with high missingness were removed. The comorbidity classes and their corresponding ICD-9/10 codes were shown in Supplementary Table S7.

We leveraged combinations of LOINC codes comprising lab tests, derived additional variables (neutrophils-to-lymphocytes ratio, albumin-to-globulin ratio, neutrophils-plus-monocytes-to-lymphocytes ratio, and platelets-to-lymphocytes ratio) and performed the following cleanup steps: 1. Removal of high missingness features (with missing rates > 80%); 2. Removal of low-variance, highly-correlated features to arrive at the labs presented for this cohort. The LOINC codes of the

lab tests and the equations to calculate the derived features were provided in Supplementary Table S8. In total, we extracted 37 lab features, derived 4 features, and following data cleanup, resulted in 24 features.

[1]. Wong, W. B. *et al.* Does adding claims to EMR data improve comorbidity burden capture? A Concerto Health AI Definitive Oncology Dataset and Symphony Linkage Study. *J. Clin. Oncol.* **37**, e18186–e18186 (2019).

[2]. Moore, B. J., White, S., Washington, R., Coenen, N. & Elixhauser, A. Identifying increased risk of readmission and in-hospital mortality using hospital administrative data: the AHRQ Elixhauser Comorbidity Index. *Medical care* **55**, 698–705 (2017).

We have revised the manuscript by adding these preprocessing details to make it clearer:

- We have revised the 5th paragraph of the “Methods-Study Setting” Section by adding the processing details on the comorbidities.
- We have revised the 6th paragraph of the “Methods-Study Setting” Section by adding the processing details on the lab tests.
- We have introduced the feature cleanup steps in the 10th paragraph of the “Methods-Study Setting” Section.

3. Missing Data Imputation: For the imputation of missing data, clarify whether the mean values were calculated based on the training data and then used to impute both the training and validation sets.

Answer: Thanks for pointing this out. The mean values were indeed calculated based on the training data and used to impute both the training and validation sets. We have revised the manuscript by clarifying it in the last paragraph of the “Methods-Study Setting” Section.

4. Comparison of Methods: Table 2 reports the comparison of methods using the validation set. Please also include a comparison of methods using the training set to demonstrate that there is no overfitting issue.

Answer: Thanks for the suggestion. We have reported the performance metrics in cross-validation on the development cohort (training set) in Supplementary Table S1. As shown in the table, the performance metrics for the development cohort are comparable to those for the validation cohort, indicating that there is no overfitting issue.

5. Vital Signs in Figure 4: Group C is described as including abnormal lab tests and vital signs, yet no vital signs appear to be included. Does this mean vital signs did not contribute to the predictive model? Please explain. Additionally, abnormal lab tests can indicate either high or low values. It would be beneficial to analyze whether each abnormality is high or low for each lab test in this figure.

Answer: The vital signs (heart rate, systolic blood pressure, diastolic blood pressure, oxygen saturation) contribute to the cardiovascular group within Figure 4 Group C. The construction of these groups (e.g., cardiovascular, renal, hematologic functions) are described under the 4th paragraph of the “Subphenotype Visualization” Section in Supplementary Methods.

Assessing direction of abnormality is an excellent suggestion. In Supplementary Table S12, we have reported the number of patients with high or low abnormalities for each lab test shown in

Figure 4. For 16 out of the 19 lab test features, more than 90% of the abnormalities are in the same direction. This observation is consistent across both the development and validation cohorts.

6. Exclusion of Age: Age was not included in the list of demographic factors that were input into the model. Please provide an explanation for its exclusion.

Answer: Thanks for the comment and we apologize for not making this clear. Age was included in the demographic factors. We provided details in Table 3 and Supplement Table S6 under demographics.

Some minor comments:

7. The limitations section should address whether the proposed framework can be applied to other disease analyses.

Answer: Thanks for the suggestion. Our proposed framework can be generalizable to other disease analyses. We have revised the manuscript by discussing this in the 4th paragraph of the Discussion Section.

8. Although the framework used the last lab test results before the index date, a discussion on the potential benefits of considering changes in lab test results for NSCLC or other diseases would be useful.

Answer: Thank you for your suggestion. Incorporating longitudinal information (e.g., trends or changes in lab test and vital sign results) could potentially enhance performance. However, most patients had ≤ 1 recorded result for these tests (58.6% on average) during the observational period in the cohort. Therefore, we leave the exploration of longitudinal information for future research using datasets with more comprehensive longitudinal data.

We have discussed this limitation of the current data in the 5th paragraph of the “Discussion” Section.

Reviewer #1 (Remarks on code availability):

Based on the information provided, the code pertains to a different manuscript titled "Generalizability of a Machine Learning Model for Improving Utilization of Parathyroid Hormone-Related Peptide Testing Across Multiple Clinical Centers." Although some portions of the code may be applicable to this submission, it would enhance reproducibility to create a separate repository specifically dedicated to this manuscript.

Answer: Thanks for catching this. The link and code were correct. We have corrected this typo in the README file.

Reviewer #3 (Remarks to the Author):

This was a comprehensive, well-done study. The proposed model was compared to multiple baselines, the figures were informative and illustrated important concepts, figures 3 and 4 make it easy to see that the phenotypes are consistent across the development

and validation set, and the methods described the cost function in detail and how the hyperparameters were selected so that the reader can be confident that you're not overhyping and an appropriate number of phenotypes were determined.

There are not any major concerns, but there are some additional details that would be nice to report.

1. I would suggest the use of Table 1 from Bos et. al.'s paper "The perils of premature phenotyping in COVID-19: a call for caution" as a checklist for mature phenotypes. I would check a box next to timing, derivation, size of derivation cohort, validation, internal coherence, relationship to underlying biology, and complexity. Impact on practice was mentioned as a limitation in the paper. Discreteness/continuity is the only row in this table that may need slightly more explanation in the discussion section. You describe how phenotype 2 is different than phenotypes 1 and 3, and you justify selecting three phenotypes based on majority voting from the results of NbClust, but it should be mentioned whether phenotype 2 is a discrete phenotype or is a continuous part of the space between phenotypes 1 and 3? It would also be informative to report the full results of NbClust; were three subphenotypes a clear winner or only slightly better than two subphenotypes?

Answer: Thanks for the suggestion. We have added Supplementary Table S10 to report the full results of NbClust. Among the clustering metrics, 10 metrics identified 3 as the optimal number of clusters, while 4 metrics favored 2 as the optimal cluster number. The results showed that the number of three is a clear winner over the number of two.

2. On line 482, the graph neural network uses the top-5 nearest neighbors. Could this have been a hyperparameter? How can you justify that the proposed method is either insensitive to this choice or that this choice is optimal?

Answer: Thanks for the question. We have added Supplementary Table S11 to compare different strategies by varying similarity thresholds and using fixed numbers of neighbors, based on performance in the development cohort. The results in the table show that using a fixed number of five neighbors yielded the best performance.

3. On line 539, it was stated that the lambda parameters could be determined empirically. Would you report the values of lambda that were used? It would be informative to know how much the loss function was weighted towards each individual component.

Answer: Thanks for the suggestion. We have reported the lambda parameters that were used in the 2nd paragraph of the "Methods-Evaluation" Section.

4. There are also a few typos to correct:

- (1). Line 517 says uncensored but you meant censored
- (2). Line 555 period should be a comma and We should be lowercase
- (3). Line 604 change my method to our proposed method
- (4). Supplementary table 3, three column headers have the word percentage split across two lines without a hyphen.

Answer: Thanks for catching these typos. We have fixed these typos in the revised manuscript.

Reviewer #4 (Remarks to the Author):

Pan et al present a graph neural network (GNN) based approach to phenotypically cluster patients with EHR data, called GEMS. They apply this approach to the concertAI database including patients with advanced NSCLC treated with immunotherapy and find, in both 5 fold cross validation and a held out cohort in the same database, that the clusters have prognostic power.

1. However, the method is not benchmarked against more straightforward clustering methods aside from a qualitative UMAP presentation in figure 2; the justification for the GNN is not clear.

Answer: Thanks for the comment. In addition to the qualitative UMAP presentation in Figure 2, we also conducted the ablation study to evaluate the impact of GNN encoder, as shown in Supplementary Table S2. We observed that the model with GNN encoder performs better than the models without an encoder or leveraging an MLP encoder.

We also implemented classical unsupervised clustering methods including K-means and Hierarchical Agglomerative Clustering (HAC) and added the results in Table 2.

2. The molecular data included seem somewhat arbitrary; several genes with known significance are excluded.

Answer: Thanks for the comment. The ConcertAI data is one of the largest and highly curated real-world datasets in oncology, with multiple publications. It compiles structured and curated EHR data originating from oncology clinics. However, as this data is primarily from the community hospital setting rather than academic centers, full molecular testing panels are limited and may not be presented in reports provided to the clinics; thus, the ConcertAI has limited capture of full tumor genomic profiling. We retained biomarker variables that had at least 100 non-null samples to come up with the variables presented in the manuscript to avoid preferential selection of variables, including genomic features. We have also acknowledged this limitation in the 5th paragraph of the “Discussion” Section and reported the missingness/positive/negative rate of all molecular features captured across the cohort as Supplementary Table S9.

3. The survival prediction results are benchmarked against standard methods but perform only marginally better if at all (confidence intervals are not provided, and the c indices are better than the next best methods by 1-2%).

Answer: Thanks for the comment. We have updated Table 2 to include the 95% confidence intervals for the survival prediction results across all methods.

4. The important features identified by the model are already known to have prognostic significance in NSCLC. Although the analyses are reasonably performed, the results appear to be of limited computational and biological significance. I have some suggestions to improve the manuscript, but to be honest, unless the results of these suggested

analyses produce something unexpectedly striking, the novelty and impact of this manuscript will likely still be quite limited.

Answer: Thank you for the comment. To clarify, our manuscript has the following contributions:

1. In terms of computational innovation, we developed a novel machine learning algorithm that can identify clusters that are associated with survival outcomes. Compared to conventional unsupervised clustering approaches which do not consider any outcomes or supervisions, our method accounts for the survival outcomes to ensure their distributions are different across different clusters. Compared to supervised methods such as mixture survival prediction which focus on the prediction performance, our approach enforces the similarity among samples within each cluster to make sure the homogeneity of their baseline characteristics so the clusters are more interpretable and can be regarded as subphenotypes.

2. In the context of aNSCLC, our algorithm uniquely identifies groups of patients with different survival curves with descriptive underlying characteristics. These characteristics (features) have been previously shown in the literature and in the clinic as having prognostic importance; thereby lending credence to our methodology. The consistency across development and validation cohorts also validated the robustness of the derived subphenotypes.

3. Although the identified important features are largely known, we analyzed the subphenotypes identified by these features to provide more comprehensive insights into the heterogeneity of aNSCLC patients. In addition to Subphenotypes 1 and 3, which correspond to subpopulations with overall mild and severe baseline conditions and outcomes, respectively, Subphenotype 2 was characterized by a distinct profile, including relatively poorer renal function and lower hematocrit levels, but better cardiovascular function.

5. Straightforward clustering on the features directly without the GNN should be added as a method for benchmarking in the survival analyses.

Answer: Thanks for the suggestion. We have implemented K-means, Hierarchical Agglomerative Clustering (HAC) and added their results in Table 2.

6. Full molecular data from tumor genomic profiling should be included. I see tumor mutational burden, MET, PTEN, and STK11 were included so this must exist for the cohort, but these 4 features seem extremely arbitrary/poorly chosen.

Answer: Thanks for restating the concern. Please see the answer of Q2.

7. The GEMS algorithm developed in the immunotherapy cohort should be applied to non-immunotherapy cohorts to test whether the clusters are predictive of differential response to immunotherapy or merely prognostic.

Answer: Thanks for the comment. This would definitely help highlight difference in clusters between IO and non-IO groups. However, we lack access to an equivalent non-immunotherapy cohort to perform this analysis. In the 5th paragraph of the “Discussion” Section., we have made it clear that this is applicable within an IO mono/combination therapy cohort only.

8. Results from the individual fold cross validation results would help readers get a sense of the variation in c indices.

Answer: Thanks for the suggestion. We reported the results of the cross validation in the development cohort in Supplementary Table S1.

9. It would be helpful to know what immunotherapy treatments patients received and in combination with what other cytotoxic or targeted agents.

Answer: Thanks for the comment. We have provided this as additional descriptive data in Table 1.

Review Response Letter

We appreciate the constructive comments and feedback from the reviewers. We greatly appreciate reviewers' positive feedback to our manuscript.

Reviewer #1 (Remarks to the Author):

The authors have thoroughly addressed my comments, revised the manuscript accordingly, and provided additional information in the supplemental materials.

Reviewer #3 (Remarks to the Author):

The authors addressed all my concerns, thank you.

Reviewer #3 (Remarks on code availability):

Code and example dataset were made available.

Reviewer #4 (Remarks to the Author):

The authors have made a reasonable attempt to address my comments given the limitations of their dataset.